

# Constructing distinct color sets of $k$ -mers via fingerprinting

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## Colored $k$ -mers

- Way to index massive bacterial datasets.
- Each genome is a color. Each  $k$ -mer is associated with a set of colors (the **color set** of the  $k$ -mer).
- Interesting object: the set of distinct color sets.
- Key to compressing the data structure.

# Color matrix

|       | $G_1$ | $G_2$ | $G_3$ | $G_4$ | $G_5$ | $G_6$ | $G_7$ | $G_8$ |
|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| ACGTA | 1     | 0     | 1     | 1     | 0     | 0     | 1     | 0     |
| CGTAC | 1     | 1     | 0     | 0     | 1     | 0     | 0     | 1     |
| GTACG | 0     | 1     | 1     | 0     | 0     | 1     | 0     | 0     |
| TACGT | 1     | 0     | 1     | 1     | 0     | 0     | 1     | 0     |
| ACGTG | 0     | 0     | 1     | 0     | 1     | 1     | 0     | 1     |
| TGCAA | 1     | 1     | 0     | 1     | 0     | 0     | 0     | 0     |
| GCAAC | 0     | 1     | 1     | 0     | 0     | 1     | 0     | 0     |
| CAACT | 0     | 0     | 0     | 1     | 1     | 0     | 1     | 1     |
| TTGCA | 1     | 0     | 1     | 1     | 0     | 0     | 1     | 0     |
| AACGT | 0     | 1     | 1     | 0     | 1     | 0     | 1     | 0     |
| CGTAT | 0     | 0     | 1     | 0     | 1     | 1     | 0     | 1     |
| GTATC | 0     | 0     | 1     | 0     | 0     | 1     | 1     | 0     |
| TATCG | 1     | 1     | 0     | 0     | 1     | 0     | 0     | 0     |
| ATCGA | 0     | 1     | 1     | 1     | 0     | 0     | 1     | 1     |
| CGAAC | 0     | 0     | 1     | 0     | 1     | 1     | 0     | 1     |

# Color matrix

|       | $G_1$ | $G_2$ | $G_3$ | $G_4$ | $G_5$ | $G_6$ | $G_7$ | $G_8$ |
|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| ACGTA | 1     | 0     | 1     | 1     | 0     | 0     | 1     | 0     |
| CGTAC | 1     | 1     | 0     | 0     | 1     | 0     | 0     | 1     |
| GTACG | 0     | 1     | 1     | 0     | 0     | 1     | 0     | 0     |
| TACGT | 1     | 0     | 1     | 1     | 0     | 0     | 1     | 0     |
| ACGTG | 0     | 0     | 1     | 0     | 1     | 1     | 0     | 1     |
| TGCAA | 1     | 1     | 0     | 1     | 0     | 0     | 0     | 0     |
| GCAAC | 0     | 1     | 1     | 0     | 0     | 1     | 0     | 0     |
| CAACT | 0     | 0     | 0     | 1     | 1     | 0     | 1     | 1     |
| TTGCA | 1     | 0     | 1     | 1     | 0     | 0     | 1     | 0     |
| AACGT | 0     | 1     | 1     | 0     | 1     | 0     | 1     | 0     |
| CGTAT | 0     | 0     | 1     | 0     | 1     | 1     | 0     | 1     |
| GTATC | 0     | 0     | 1     | 0     | 0     | 1     | 1     | 0     |
| TATCG | 1     | 1     | 0     | 0     | 1     | 0     | 0     | 0     |
| ATCGA | 0     | 1     | 1     | 1     | 0     | 0     | 1     | 1     |
| CGAAC | 0     | 0     | 1     | 0     | 1     | 1     | 0     | 1     |

|       | $G_1$ | $G_2$ | $G_3$ | $G_4$ | $G_5$ | $G_6$ | $G_7$ | $G_8$ | Fingerprint       |
|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------------------|
| ACGTA | 1     | 0     | 1     | 1     | 0     | 0     | 1     | 0     | <b>1011010011</b> |
| CGTAC | 1     | 1     | 0     | 0     | 1     | 0     | 0     | 1     | <b>0110100101</b> |
| GTACG | 0     | 1     | 1     | 0     | 0     | 1     | 0     | 0     | <b>1100011110</b> |
| TACGT | 1     | 0     | 1     | 1     | 0     | 0     | 1     | 0     | <b>1011010011</b> |
| ACGTG | 0     | 0     | 1     | 0     | 1     | 1     | 0     | 1     | <b>0101110100</b> |
| TGCAA | 1     | 1     | 0     | 1     | 0     | 0     | 0     | 0     | <b>1001001010</b> |
| GCAAC | 0     | 1     | 1     | 0     | 0     | 1     | 0     | 0     | <b>1100011110</b> |
| CAACT | 0     | 0     | 0     | 1     | 1     | 0     | 1     | 1     | <b>0010111001</b> |
| TTGCA | 1     | 0     | 1     | 1     | 0     | 0     | 1     | 0     | <b>1011010011</b> |
| AACGT | 0     | 1     | 1     | 0     | 1     | 0     | 1     | 0     | <b>1110001100</b> |
| CGTAT | 0     | 0     | 1     | 0     | 1     | 1     | 0     | 1     | <b>0101110100</b> |
| GTATC | 0     | 0     | 1     | 0     | 0     | 1     | 1     | 0     | <b>0011000111</b> |
| TATCG | 1     | 1     | 0     | 0     | 1     | 0     | 0     | 0     | <b>1000110010</b> |
| ATCGA | 0     | 1     | 1     | 1     | 0     | 0     | 1     | 1     | <b>0100101011</b> |
| CGAAC | 0     | 0     | 1     | 0     | 1     | 1     | 0     | 1     | <b>0101110100</b> |

| k-mer | Fingerprint       |
|-------|-------------------|
| ACGTA | <b>1011010011</b> |
| CGTAC | <b>0110100101</b> |
| GTACG | <b>1100011110</b> |
| TACGT | <b>1011010011</b> |
| ACGTG | <b>0101110100</b> |
| TGCAA | <b>1001001010</b> |
| GCAAC | <b>1100011110</b> |
| CAACT | <b>0010111001</b> |
| TTGCA | <b>1011010011</b> |
| AACGT | <b>1110001100</b> |
| CGTAT | <b>0101110100</b> |
| GTATC | <b>0011000111</b> |
| TATCG | <b>1000110010</b> |
| ATCGA | <b>0100101011</b> |
| CGAAC | <b>0101110100</b> |

Color-set covering subset of k-mers

| k-mer | Fingerprint       |
|-------|-------------------|
| CAACT | <b>0010111001</b> |
| GTATC | <b>0011000111</b> |
| ATCGA | <b>0100101011</b> |
| ACGTG | <b>0101110100</b> |
| CGTAC | <b>0110100101</b> |
| TATCG | <b>1000110010</b> |
| TGCAA | <b>1001001010</b> |
| ACGTA | <b>1011010011</b> |
| GTACG | <b>1100011110</b> |
| AACGT | <b>1110001100</b> |

# Fingerprinting scheme: Zobrist hashing

- **Precomputation:** For each color, pick an  $\ell$ -bit fingerprint uniformly at random. Denote the fingerprint of color  $c$  with  $f(c)$ .
- **Fingerprint of a set:** The fingerprint of a set  $A = \{c_1, c_2, \dots, c_m\}$  will be  $F(A) = f(c_1) \oplus f(c_2) \oplus \dots \oplus f(c_m)$ .
- **Wishlist:** Incremental ✓, Commutative ✓, Atomically updatable ✓, Collision-resistant: ?

# Collision analysis

The fingerprint function  $F$  is a **universal hash family** over the single-color fingerprint picks  $f(c)$ . By the union bound:

**Lemma 2.** *Given a set of distinct sets  $A_0, \dots, A_{N-1}$ , the probability that there exists two sets  $A_i \neq A_j$  such that  $F(A_i) = F(A_j)$  is at most  $\frac{N^2}{2^{\ell+1}}$ , where  $\ell$  is the length of a fingerprint.*

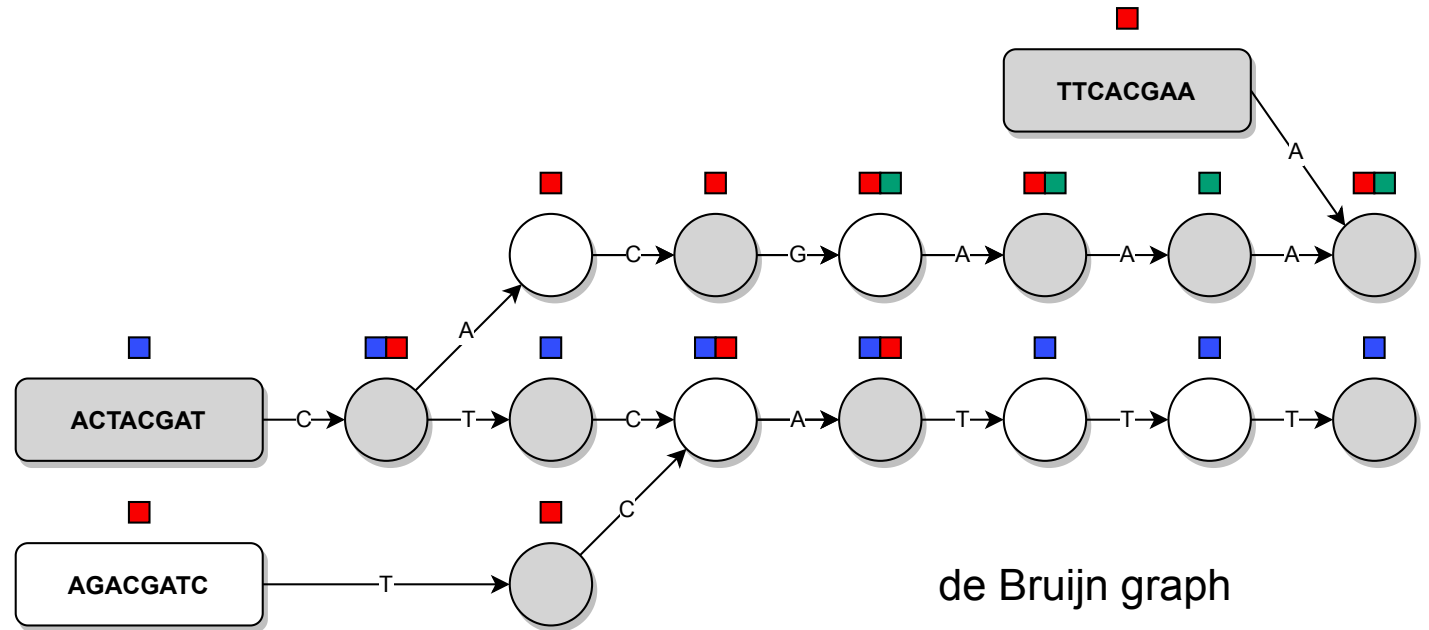
For example, for  $\ell = 128$  and  $N = 10^9$ , we have a collision probability of at most  $10^{18} / 2^{129} \approx 1.47 \cdot 10^{-21} \approx \underline{\text{don't worry about it.}}$

Okay, but storing a fingerprint for every k-mer is still a lot of memory

Adjacent  $k$ -mers often share the same color set.

Input

|                 |
|-----------------|
| ACTACGATCTCATTT |
| CTACGATCACGA    |
| TTCACGAAA       |
| AGACGATCTCA     |
| CGATCACGAAA     |



# Key $k$ -mers

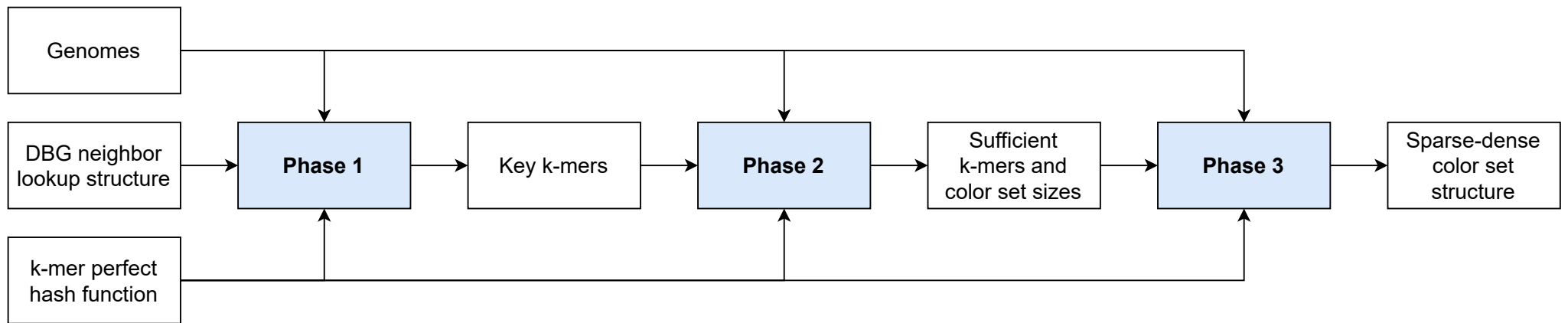
- The color set can change only when:
  - The graph branches.
  - An input sequence ends or starts.
- If one of the above applies to a  $k$ -mer, we call it a *key  $k$ -mer*.
- The key  $k$ -mers are a **color-set covering set** of  $k$ -mers, but not *minimal*.

# Algorithm

We follow a two-step filtering approach from  $k$ -mers to key  $k$ -mers, to *sufficient*  $k$ -mers.

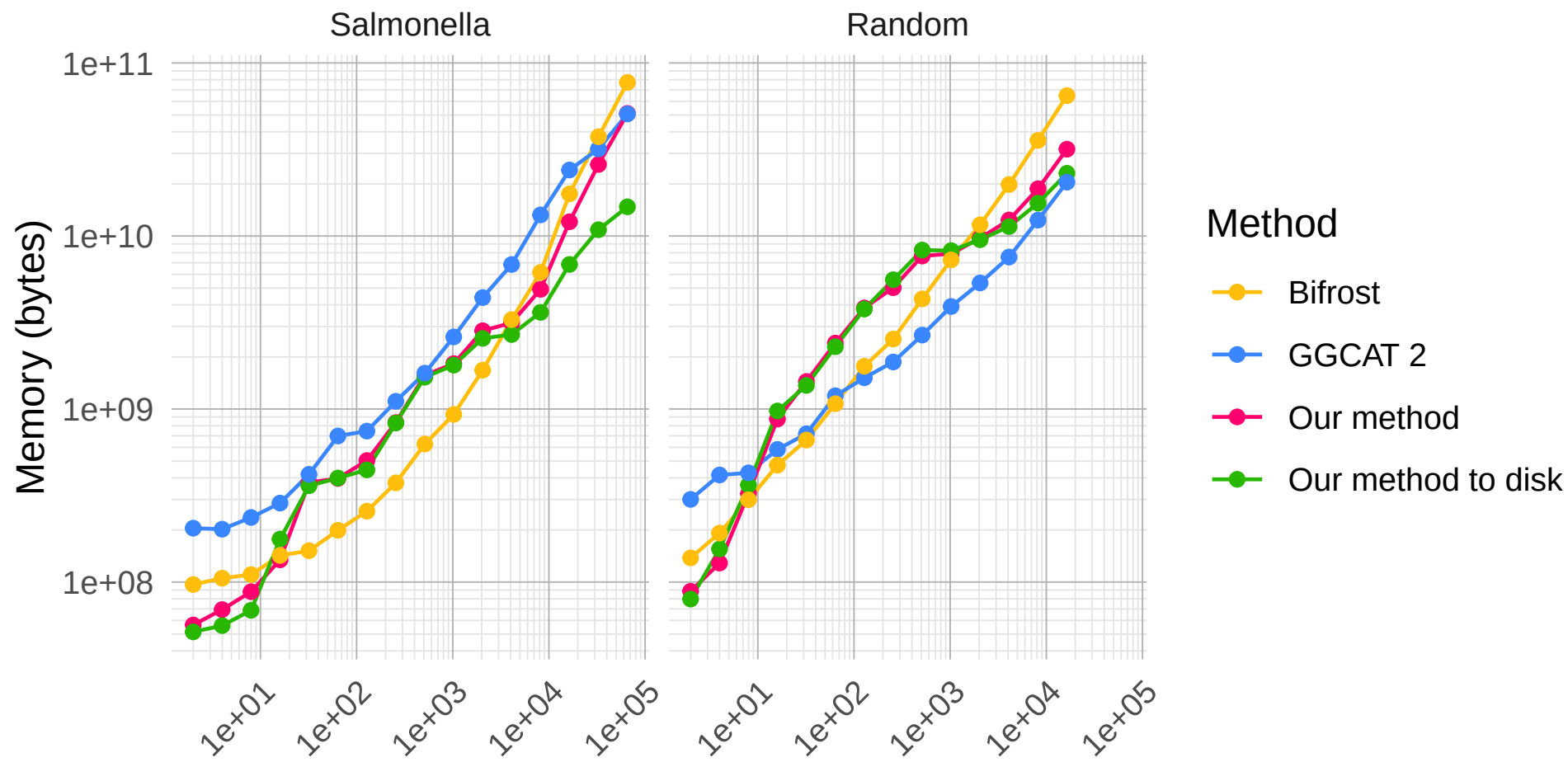
1. Find the key  $k$ -mers
2. Build color set fingerprints of the key  $k$ -mers.
3. Deduplicate the fingerprints  $\rightarrow$  sufficient  $k$ -mers.

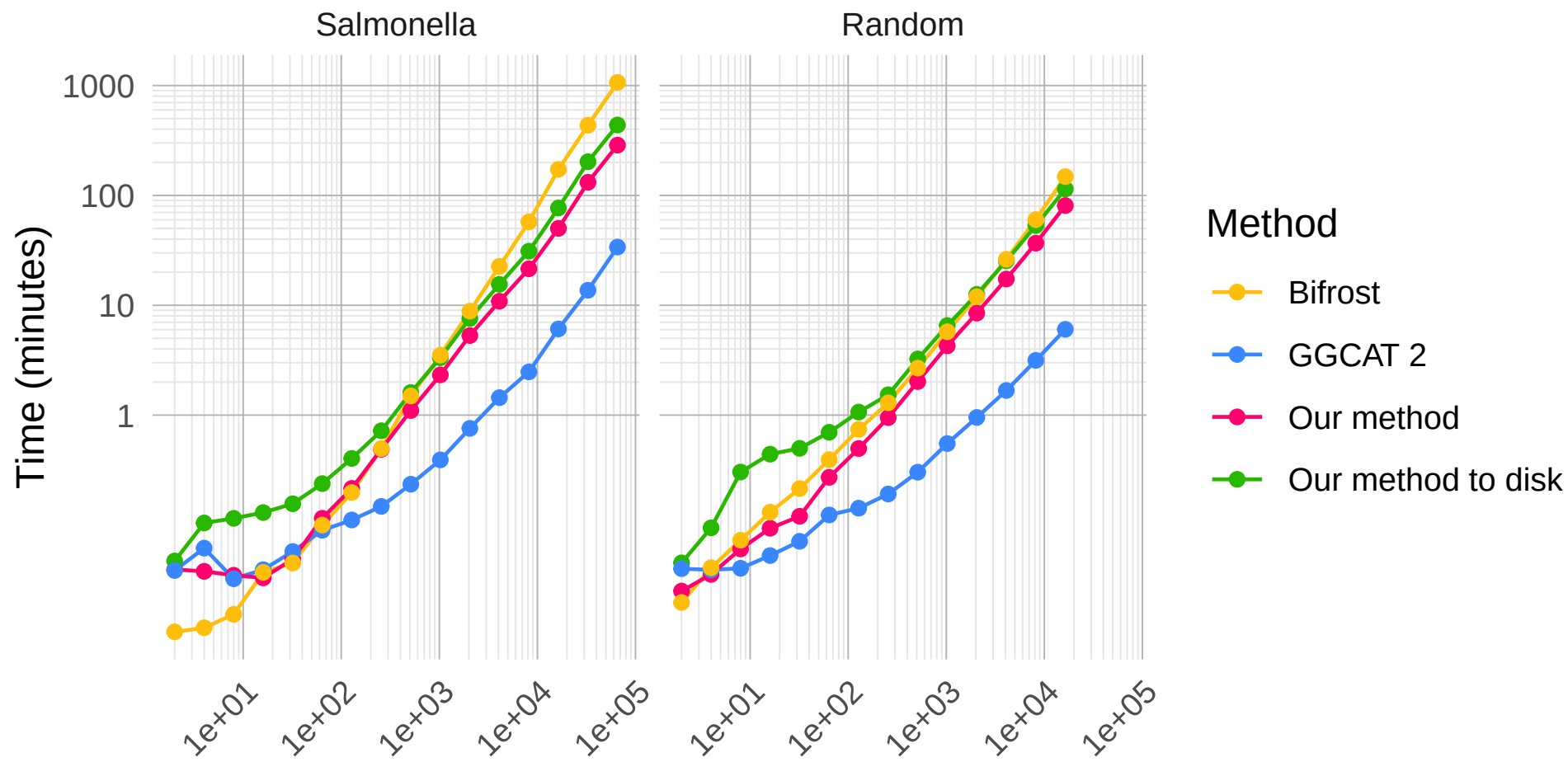
Finally, we build one color set for each distinct fingerprint, directly into a **compressed form**.

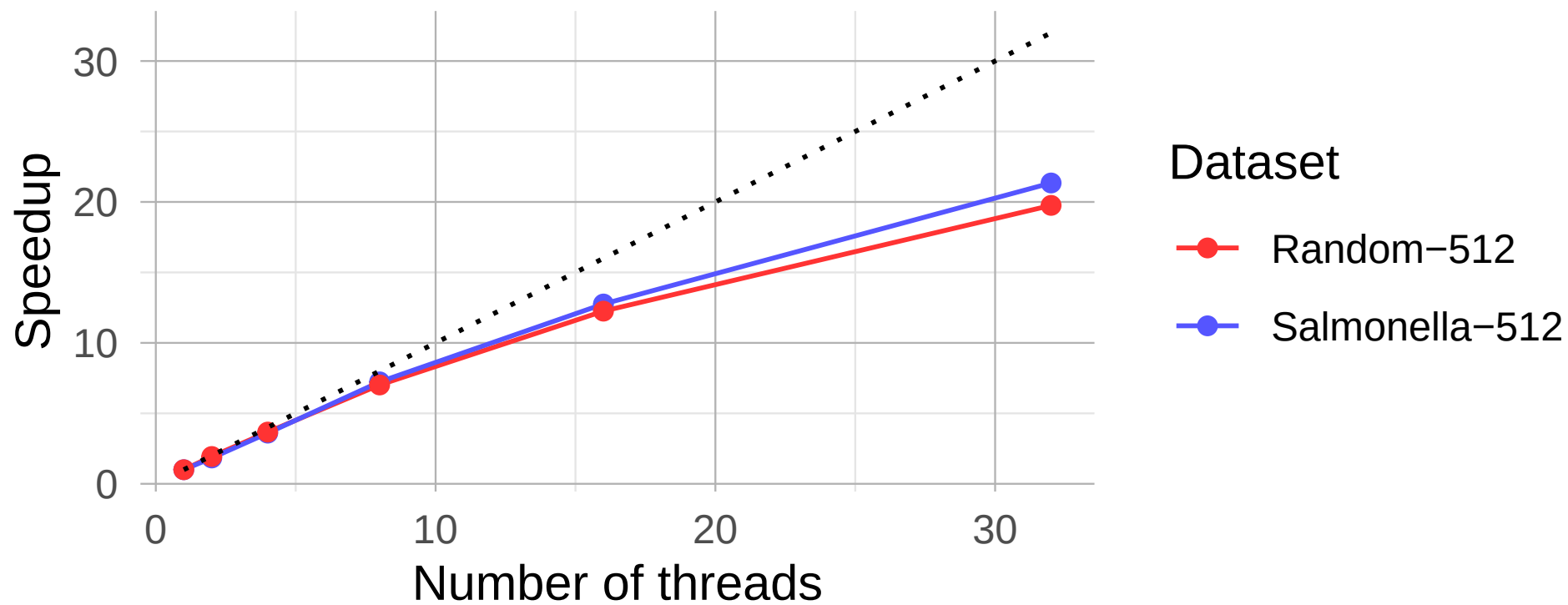


# Results

- **Machine:** 32-core Threadripper PRO 3975WX, 504 GiB RAM, 400MB/s hard drive IO.
- **Datasets:** Subsets of AllTheBacteria: 1,2,4...,65536 *Salmonella* genomes, 1,2,4,...,16384 random genomes.
- **Implementation details:**
  - SBWT was used as the perfect hash function on  $k$ -mers and for the dBg neighbor lookup.
  - Options to build fully in memory, or by writing the final structure to disk in pieces.
  - Rust 🦀.
- **Baselines:** Bifrost, GGCAT 2.







# Conclusion

Our method provides:

- Deduplication of color sets even across unitigs.
- Independent parallel threads.
- Low RAM.
- Construction directly into sparse-dense form.

Future work:

- Alternative implementations, e.g. based on Sshash.
- Better compression in the final form.